

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

The previous revision's stop signal holds for ordinary objects and does not generalise past them. **High area-to-mass objects are the easy case, not the hard one.** Measured across a thousandfold span, the estimator's fractional precision scales exactly as $1/\text{signal}$, and every high-area property set in this tree sits 8 to 305 times above GNSS. Tumbling either integrates out or announces itself rather than corrupting the estimate. All of that is synthetic. The engagement now turns on one thing that is not: **a single real high area-to-mass object with a precise ephemeris.**

What reversed, and what did not

Revision 6 reported that block separation fails outside a narrow corner of arc length and noise. That result stands exactly as written — and its terms were **ordinary objects under ordinary tracking**. It does not extend past them, because the threshold scales with the strength of the radiation-pressure signal, and high area-to-mass objects have far more of it.

at 10 m position noise, separation requires	$A C_R/m$ above
on a 22-hour arc	0.033 m ² /kg
on a 6-hour arc	0.95 m ² /kg
GNSS — what was validated	0.021 fails
high area-to-mass property sets in this source	0.16–6.3 passes by 5–190×

This was measured, not extrapolated. A thousandfold span of objects was synthesised and truth arcs generated for each. Absolute scatter came out constant to four significant figures across the whole span, with the estimate unbiased to six — so fractional error is exactly proportional to $1/\text{signal}$, and scatter is exactly linear in noise. A real population spanning only $1.44\times$ could never have supported that claim; synthesising one could.

The property sets themselves are evidence about intent. Someone built entries spanning 8 to 305 times the GNSS value into this library. Two of those entries carry mislabelled comments — one is duplicated with an unreachable second definition — so the inference rests on the values rather than the names, but the values are unambiguous about the population being targeted.

Tumbling: flagged, not fooled

The estimator assumes the parameter is constant over the arc, which for an irregular or tumbling object it is not. This was the strongest objection to the case above, and it does not hold.

Fast tumbling is harmless. At periods of an hour or less the orbit integrates it out; the fit recovers the mean to six significant figures with a residual indistinguishable from a static object.

Slow variation biases boundedly and announces itself. A $\pm 30\%$ variation over six hours biases the estimate by 3.28% — and moves the fit residual from 0.0005 m to 0.71 m. The condition is detectable rather than silent, and the signature grows with signal while the bias does not, so it is most visible on exactly the objects where the estimate is most precise.

An instrument that flags a varying effective area is more useful than one that assumes constancy, because the variation is itself information about the object.

The limit of this evidence

All of the above is synthetic, and the truth arcs were generated with the same force model that then fitted them. That is a fair test of the *estimator* and an unfair test of the *physics*, and the gap is larger than it first appears.

In the one real-data case, the binding limit was not observation noise. The formal parameter uncertainty was optimistic by 10–20× against empirical scatter, with residuals systematic and correlated along the arc — the signature of unmodelled dynamics. The GNSS result was limited by model error, not measurement error. The synthetic study varied noise, so the 1/signal scaling is established for the noise term and not for the one that actually bound the real case.

Nor does model error behave uniformly. An error independent of radiation pressure — gravity, drag — does shrink against a stronger signal. An error that *scales with* radiation pressure — wrong reflectivity, wrong shape, unknown attitude — keeps a constant fractional size however strong the signal becomes. The 3.28% tumbling bias above is exactly such a term.

So the honest form of the claim is: precision improves as 1/signal down to a floor set by how well the object's shape and attitude are known, and that floor does not improve. Whether it bites depends on the application — for separating classes that span orders of magnitude, a few percent almost certainly does not.

Two narrower caveats: for objects in low orbit, drag and radiation pressure are close to degenerate since both scale with area over mass, which is largely moot in geostationary orbit where the debris of interest mostly sits. And nothing here has been tested against an object whose shape and attitude are genuinely unknown — which is the defining property of the target population.

Where the asset stands

Rev. 1	The ray-traced SRP engine — scarce, missing its fuel
Rev. 2	The validation — a credential making the SRP claim falsifiable
Rev. 3	A parameter-estimating orbit determination capability
Rev. 4	An estimator that may characterise objects from dynamics alone
Rev. 5	It does — for large, well-tracked objects
Rev. 6	And only for those; the commercial regime looked out of reach
Rev. 7	The commercial regime is the easy one. It is demonstrated in simulation and untested on a real object, and that single test is what the question now rests on.

Seven revisions, each moved by evidence rather than by rearranging the framing. Three moved against the thesis, including the one immediately before this. That record is why the remaining claims can be put in front of someone.

Valuation

- **As an orbit determination product: approximately £0.** Unchanged across all seven revisions.
- **As a high area-to-mass characterisation capability: unproven, and the only thesis that could carry a business.** It is one real object away from being testable, and the economics favour finding that object over any further engineering.
- **As a validated body of work: the durable result.** A tree that did not build is under version control with documented provenance, an honest accuracy figure against public reference data,

a characterised estimator with measured scaling behaviour, and **five silent-wrong-answer defects found and fixed**. That last is worth stating plainly: this codebase's failure mode was consistently a confident wrong answer rather than an error, and that is the single most important thing for anyone inheriting it to know.

- **Replacement cost remains the wrong frame** at £300k–£600k.

Recommendation

The engineering is finished. One diagnostic is running and then the work stops. What remains is not a development question.

1. **Obtain one real high area-to-mass object with a precise ephemeris.** This is the whole question now, and it is a data-access problem rather than an engineering one. It is currently blocked on a credentialed download.
2. **If it confirms the synthetic result:** the single remaining investment is observation models and covariance — what converts a tool that fits published ephemerides into one that characterises objects from real tracking. Do not start it before the IP position is settled.
3. **If it does not, or if the data cannot be obtained:** stop and publish. Recovering spacecraft hardware classes from dynamics alone is an interesting result on its own terms, and finishing there is a legitimate outcome rather than a failure.
4. **Settle the IP position regardless.** Outstanding since revision 1, it gates publication as surely as sale, and it needs no engineering at all.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026 by the `odl` maintainer session, reviewed and spot-verified here. The signal-scaling and tumbling results are synthetic, as described above. The competitive and valuation sections are judgement. Revision 7 supersedes revision 6, whose negative result was correct but narrower in scope than it appeared.